Variational inference

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EM algorithm

For a latent-variable model,

$$p_\theta(x, z) = p_\theta(z)p_\theta(x \mid z), \qquad p_\theta(z \mid x) = \frac{p_\theta(x, z)}{\int p_\theta(x, u)\, du}.$$

EM averages over the current posterior:

$$Q(\theta \mid \theta^{(m)}) = \mathbb{E}_{p_{\theta^{(m)}}(z|x)}[\log p_\theta(x, Z)].$$

The M-step holds that distribution fixed:

$$\theta^{(m+1)} \in \arg\max_\theta Q(\theta \mid \theta^{(m)}).$$

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What if the posterior expectations are difficult to compute?

A nonlinear decoder

$$Z \sim \mathcal{N}(0, I), \qquad X \mid Z \sim \mathcal{N}\{f_\theta(Z), \sigma^2 I\}.$$

Its posterior satisfies

$$p_\theta(z \mid x) \propto \exp\left\{-\frac{1}{2} z^\top z - \frac{\|x - f_\theta(z)\|^2}{2\sigma^2}\right\}.$$

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For which forms of f_θ is the posterior distribution $p_\theta(z \mid x)$ Gaussian?

An easy joint density can still give a difficult posterior.

Posterior approximation

Choose a tractable family of distributions

$$\mathcal{Q} = \{q_\lambda(z) : \lambda \in \Lambda\}.$$

Our goal is to approximate the posterior $p_\theta(z \mid x)$ with a distribution in $\mathcal{Q}$.

Using **KL divergence** to measure the discrepancy, we solve

$$\boxed{q^\star \in \arg\min_{q \in \mathcal{Q}} \mathrm{KL}\{q(z) \| p_\theta(z \mid x)\}.}$$

- θ: generative model parameters, held fixed for now.
- λ: variational parameters, optimized to approximate the posterior.

KL divergence

The **Kullback–Leibler (KL) divergence** measures the discrepancy between two distributions:

$$\boxed{\mathrm{KL}(q\|r) = \mathbb{E}_q\left[\log \frac{q(Z)}{r(Z)}\right] = \int q(z) \log \frac{q(z)}{r(z)}\, dz.}$$

- **Nonnegative:** $\mathrm{KL}(q\|r) \geq 0$.

$$-\mathrm{KL}(q\|r) = \mathbb{E}_q\left[\log \frac{r(Z)}{q(Z)}\right] \leq \log \mathbb{E}_q\left[\frac{r(Z)}{q(Z)}\right] = 0.$$

- $\mathrm{KL}(q\|r) = 0$ if and only if $q = r$ almost everywhere.
- **Direction matters:** generally, $\mathrm{KL}(q\|r) \neq \mathrm{KL}(r\|q)$.

Find the equivalence

By Bayes' rule,

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)}.$$

Substituting into the KL divergence gives

$$\begin{aligned}
\mathrm{KL}\{q \| p_\theta(z \mid x)\} &= \mathbb{E}_q\left[\log \frac{q(Z)}{p_\theta(Z \mid x)}\right] \\
&= \mathbb{E}_q[\log q(Z)] - \mathbb{E}_q[\log p_\theta(x, Z)] + \underbrace{\log p_\theta(x)}_{\text{constant in } q}.
\end{aligned}$$

For fixed θ, minimizing the KL is therefore equivalent to

$$\max_{q \in \mathcal{Q}} \left\{\mathbb{E}_q[\log p_\theta(x, Z)] - \mathbb{E}_q[\log q(Z)]\right\}.$$

Deriving the ELBO

Write the evidence as an expectation under q:

$$p_\theta(x) = \int q(z) \frac{p_\theta(x, z)}{q(z)} \, dz = \mathbb{E}_q\left[\frac{p_\theta(x, Z)}{q(Z)}\right].$$

Since log is concave, **Jensen's inequality** gives

$$\begin{aligned}
\log p_\theta(x) &= \log \mathbb{E}_q\left[\frac{p_\theta(x, Z)}{q(Z)}\right] \\
&\geq \mathbb{E}_q\left[\log \frac{p_\theta(x, Z)}{q(Z)}\right] \\
&= \mathbb{E}_q[\log p_\theta(x, Z)] - \mathbb{E}_q[\log q(Z)] =: \mathcal{L}(q, \theta).
\end{aligned}$$

$\mathcal{L}(q, \theta)$ is the evidence lower bound (ELBO): $\mathcal{L}(q, \theta) \leq \log p_\theta(x)$.

The posterior KL gap

The ELBO identity shows exactly how far the bound is from the log evidence:

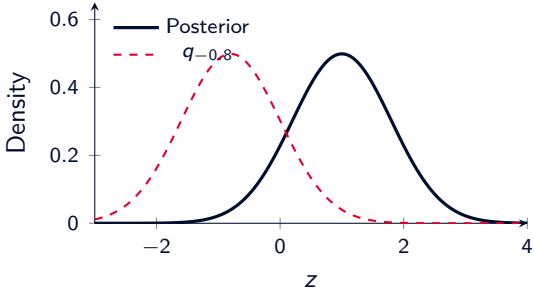
$$\log p_\theta(x) = \mathcal{L}(q, \theta) + \underbrace{\mathrm{KL}\{q \| p_\theta(z \mid x)\}}_{\text{gap} \geq 0}.$$

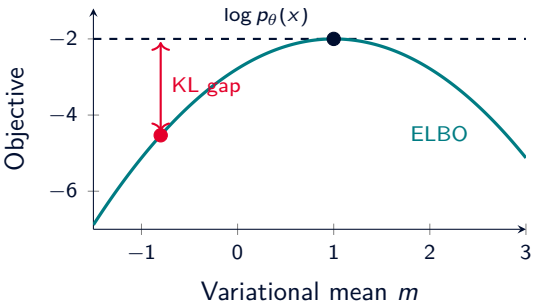
- **For fixed θ:** maximizing the ELBO is equivalent to minimizing the KL divergence from q to the posterior.
- **When is the bound tight?**

$$\mathcal{L}(q, \theta) = \log p_\theta(x) \quad \Longleftrightarrow \quad q(z) = p_\theta(z \mid x) \quad \text{a.e.}$$

A picture of the ELBO

$$p_\theta(z \mid x) = \mathcal{N}(1, 0.8^2), \qquad q_m = \mathcal{N}(m, 0.8^2).$$





For illustration, set $\log p_\theta(x) = -2$. Then

$$\mathcal{L}(q_m, \theta) = -2 - \underbrace{\frac{(m-1)^2}{1.28}}_{\text{posterior KL gap}}.$$

ELBO interpretation I

Recall the definition of the ELBO:

$$\mathcal{L}(q, \theta) = \mathbb{E}_q[\log p_\theta(x, Z)] - \mathbb{E}_q[\log q(Z)].$$

Define the **entropy** of q as

$$\mathcal{H}(q) = -\mathbb{E}_q[\log q(Z)].$$

Then

$$\boxed{\mathcal{L}(q, \theta) = \underbrace{\mathbb{E}_q[\log p_\theta(x, Z)]}_{\text{expected log joint}} + \underbrace{\mathcal{H}(q)}_{\text{entropy}}.}$$

- **Expected log joint:** favors maximum a posteriori (MAP) estimate.
- **Entropy:** rewards a more spread-out approximation.

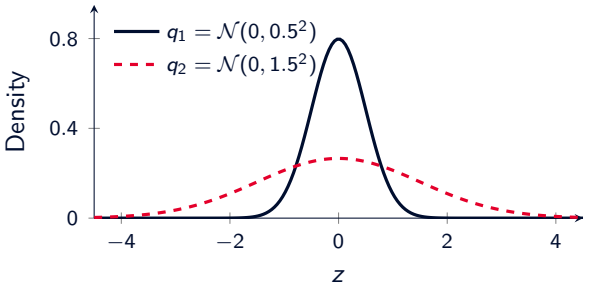
What entropy measures

Entropy measures uncertainty in a draw from a distribution:

$$\mathcal{H}(q) = -\mathbb{E}_q[\log q(Z)].$$

For $q = \mathcal{N}(\mu, \sigma^2)$,

$$\boxed{\mathcal{H}(q) = \frac{1}{2}\log(2\pi e\sigma^2).}$$



Narrow Gaussian:

$$\sigma = 0.5 \quad \Rightarrow \quad \mathcal{H}(q_1) \approx 0.73.$$

Wide Gaussian:

$$\sigma = 1.5 \quad \Rightarrow \quad \mathcal{H}(q_2) \approx 1.82.$$

ELBO interpretation II

Using $p_\theta(x, z) = p_\theta(x \mid z)p_\theta(z)$,

$$\mathcal{L}(q, \theta) = \mathbb{E}_q[\log p_\theta(x \mid Z)] + \mathbb{E}_q[\log p_\theta(Z)] - \mathbb{E}_q[\log q(Z)]$$

$$= \mathbb{E}_q[\log p_\theta(x \mid Z)] - \mathbb{E}_q\left[\log \frac{q(Z)}{p_\theta(Z)}\right].$$

Therefore,

$$\boxed{\mathcal{L}(q, \theta) = \underbrace{\mathbb{E}_q[\log p_\theta(x \mid Z)]}_{\text{expected log likelihood}} - \underbrace{\mathrm{KL}\{q(z) \| p_\theta(z)\}}_{\text{prior KL}}.}$$

- **Expected log likelihood:** rewards latent values that explain the observed x.
- **Prior KL:** penalizes departures of q from the prior.

Variational EM

For independent observations, let $q_{\lambda_i}(z_i)$ approximate $p_\theta(z_i \mid x_i)$. The total ELBO is

$$\mathcal{L}(\theta, \lambda_{1:n}) = \sum_{i=1}^{n} \left\{ \mathbb{E}_{q_{\lambda_i}}[\log p_\theta(x_i, Z_i)] + \mathcal{H}(q_{\lambda_i}) \right\}.$$

Variational E-step: update the posterior approximations.

$$\lambda_i^{(t+1)} \in \arg\max_{\lambda_i} \mathcal{L}_i(\theta^{(t)}, \lambda_i), \qquad i = 1, \ldots, n.$$

M-step: update the model parameters.

$$\theta^{(t+1)} \in \arg\max_{\theta} \sum_{i=1}^{n} \mathbb{E}_{q_{\lambda_i^{(t+1)}}}[\log p_\theta(x_i, Z_i)].$$

Exact maximization is too expensive: each step only needs to increase its objective.

Why variational EM works

Variational EM performs **coordinate ascent on the ELBO**. Write $\lambda = (\lambda_1, \ldots, \lambda_n)$.

$$\begin{aligned} \mathcal{L}(\theta^{(t)}, \lambda^{(t)}) &\leq \mathcal{L}(\theta^{(t)}, \lambda^{(t+1)}) && \text{(E-step)} \\ &\leq \mathcal{L}(\theta^{(t+1)}, \lambda^{(t+1)}) && \text{(M-step).} \end{aligned}$$

The M-step increases the same ELBO because the entropy terms $\mathcal{H}(q_{\lambda_i^{(t+1)}})$ are constant with respect to θ.

Each iteration increases the ELBO.

- With approximate posteriors, the observed log likelihood need not increase.
- If the E-step finds the true posteriors, we recover ordinary EM and its likelihood ascent guarantee.

Example 1

Generative model (linear Gaussian decoder):

$$Z_i \sim \mathcal{N}(0, I_K), \qquad X_i \mid Z_i \sim \mathcal{N}(WZ_i, I_D),$$

where $W \in \mathbb{R}^{D \times K}$ is the parameter to estimate.

Variational approximation:

$$q_i(z_i) = \mathcal{N}(\mu_i, \Sigma_i), \qquad \Sigma_i \succ 0.$$

Fit a separate approximation for each observation.

The ELBO for subject i:

$$\begin{aligned}\mathcal{L}_i &= \mathbb{E}_{q_i}[\log p_\theta(x \mid Z)] - \mathrm{KL}\{q_i \| \mathcal{N}(0, I_K)\} \\ &= \mathbb{E}_{q_i}\left[-\frac{D}{2}\log(2\pi) - \frac{1}{2}\|x_i - WZ_i\|^2\right] - \mathrm{KL}\{q_i \| \mathcal{N}(0, I_K)\}\end{aligned}$$

ELBO

1. Expected reconstruction error. Using $\mathbb{E}_{q_i}[Z_i] = \mu_i$ and $\mathbb{E}_{q_i}[Z_i Z_i^\top] = \Sigma_i + \mu_i \mu_i^\top$,

$$\begin{aligned}\mathbb{E}_{q_i} \|x_i - W Z_i\|^2 &= x_i^\top x_i - 2 x_i^\top W \mu_i + \operatorname{tr}\{W^\top W(\Sigma_i + \mu_i \mu_i^\top)\} \\ &= \|x_i - W\mu_i\|^2 + \operatorname{tr}(W^\top W \Sigma_i).\end{aligned}$$

2. KL divergence to the Gaussian prior.

$$\mathrm{KL}\{q_i \| \mathcal{N}(0, I_K)\} = \frac{1}{2}\left\{\operatorname{tr}(\Sigma_i) + \|\mu_i\|^2 - K - \log\det \Sigma_i\right\}.$$

3. Combine the two terms.

$$\boxed{\begin{aligned}\mathcal{L}_i = &-\frac{D}{2}\log(2\pi) - \frac{1}{2}\|x_i - W\mu_i\|^2 - \frac{1}{2}\operatorname{tr}(W^\top W \Sigma_i) \\ &- \frac{1}{2}\left\{\operatorname{tr}(\Sigma_i) + \|\mu_i\|^2 - K - \log\det \Sigma_i\right\}.\end{aligned}}$$

The variational E-step

Hold W fixed and define

$$A = I_K + W^\top W \succ 0, \qquad b_i = W^\top x_i.$$

Collecting terms in (μ_i, Σ_i) gives

$$\mathcal{L}_i = C_i + b_i^\top \mu_i - \frac{1}{2}\mu_i^\top A \mu_i - \frac{1}{2}\operatorname{tr}(A\Sigma_i) + \frac{1}{2}\log\det\Sigma_i,$$

where C_i does not depend on μ_i or Σ_i.

Differentiate and set both gradients to zero:

$$\nabla_{\mu_i}\mathcal{L}_i = b_i - A\mu_i = 0 \qquad \Longrightarrow \qquad \boxed{\mu_i^\star = A^{-1}W^\top x_i},$$

$$\nabla_{\Sigma_i}\mathcal{L}_i = \frac{1}{2}(\Sigma_i^{-1} - A) = 0 \qquad \Longrightarrow \qquad \boxed{\Sigma_i^\star = A^{-1}}.$$

- The optimal mean depends on x_i; the optimal covariance is shared across observations.

The true posterior

Complete the square:

$$\begin{aligned}\log p_W(x_i, z_i) &= C - \frac{1}{2} z_i^\top A z_i + b_i^\top z_i \\ &= C' - \frac{1}{2}(z_i - A^{-1} b_i)^\top A (z_i - A^{-1} b_i).\end{aligned}$$

Thus

$$p_W(z_i \mid x_i) = \mathcal{N}(A^{-1} b_i, A^{-1}) = q_i^\star(z_i).$$

VI is exact because the family contains the posterior.

The exact M-step

Hold the variational means and covariances fixed:

$$\nabla_W \mathcal{L} = \sum_i \{x_i \mu_i^\top - W(\Sigma_i + \mu_i \mu_i^\top)\}.$$

Set this gradient to zero:

$$\boxed{W^{\text{new}} = \left(\sum_i x_i \mu_i^\top\right)\left(\sum_i [\Sigma_i + \mu_i \mu_i^\top]\right)^{-1}.}$$

The M-step also uses the latent covariance.

Why VI can be expensive

The ELBO involves both a sum over observations and expectations over latent variables:

$$\mathcal{L}(\theta, \lambda_{1:n}) = \sum_{i=1}^{n} \left\{ \mathbb{E}_{q_{\lambda_i}}[\log p_\theta(x_i, Z_i)] + \mathcal{H}(q_{\lambda_i}) \right\}.$$

Computational bottleneck	**Potential solution**
Optimize separate variational parameters for every observation, including new data.	
Evaluate a sum over all n observations at every parameter update.	
Evaluate latent-variable expectations that may have no closed form.	

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Evaluate a sum over all n observations at every parameter update.	**Stochastic VI:** estimate the full-data gradient using a random minibatch.
Evaluate latent-variable expectations that may have no closed form.	**Monte Carlo:** approximate expectations using samples from the variational distribution.

Amortized VI

So far, we have optimized separate variational parameters λ_i for each observation x_i.

Amortized VI learns a shared function, called an **encoder**, that maps observations to variational parameters:

$$\lambda_i = g_\phi(x_i), \qquad q_\phi(z \mid x_i) = q_{g_\phi(x_i)}(z).$$

- **Shared parameters:** train one set of encoder weights ϕ using the ELBO across observations.
- **Observation-specific distributions:** different inputs x_i produce different variational parameters λ_i.

Example 2: VAE

Generative model (decoder):

$$Z_i \sim \mathcal{N}(0, I_K), \qquad X_i \mid Z_i \sim \mathcal{N}(f_\theta(Z_i), I_D),$$

where f_θ is a neural network mapping latent variables to the mean of the observations.

Amortized posterior approximation (encoder):

$$q_\phi(z_i \mid x_i) = \mathcal{N}(\mu_\phi(x_i), \Sigma_\phi(x_i)), \qquad \Sigma_\phi(x_i) \succ 0.$$

The encoder produces the variational parameters:

$$\boxed{\lambda_i = g_\phi(x_i) = (\mu_\phi(x_i), \Sigma_\phi(x_i)).}$$

- θ is shared across observations and controls the decoder.
- ϕ is shared across observations, but the posterior mean and covariance depend on x_i.

Training amortized VI

Learn the model and the shared inference rule by maximizing the total ELBO:

$$\max_{\theta,\phi} \mathcal{L}(\theta,\phi) = \max_{\theta,\phi} \sum_{i=1}^{n} \mathcal{L}_i(\theta,\phi).$$

For each observation,

$$\mathcal{L}_i(\theta,\phi) = \underbrace{\mathbb{E}_{q_\phi(z|x_i)}[\log p_\theta(x_i \mid Z)]}_{\text{expected log likelihood}} - \underbrace{\mathrm{KL}\{q_\phi(z \mid x_i) \| p(z)\}}_{\text{prior KL}}.$$

Stochastic VI

The total ELBO is $\mathcal{L}(\theta, \phi) = \sum_{i=1}^{n} \mathcal{L}_i(\theta, \phi)$. At each iteration, sample a uniform minibatch $\mathcal{B}$ of size B and compute

$$\boxed{\widehat{\mathcal{L}}_{\mathcal{B}}(\theta, \phi) = \frac{n}{B} \sum_{i \in \mathcal{B}} \mathcal{L}_i(\theta, \phi).}$$

The scaling n/B gives an **unbiased estimate**:

$$\mathbb{E}_{\mathcal{B}}[\widehat{\mathcal{L}}_{\mathcal{B}}] = \mathcal{L}, \qquad \mathbb{E}_{\mathcal{B}}[\nabla_{\theta,\phi} \widehat{\mathcal{L}}_{\mathcal{B}}] = \nabla_{\theta,\phi} \mathcal{L}.$$

Stochastic gradient ascent: with step size $\rho_t > 0$,

$$\theta^{(t+1)} = \theta^{(t)} + \rho_t \nabla_{\theta} \widehat{\mathcal{L}}_{\mathcal{B}_t}(\theta^{(t)}, \phi^{(t)}),$$

$$\phi^{(t+1)} = \phi^{(t)} + \rho_t \nabla_{\phi} \widehat{\mathcal{L}}_{\mathcal{B}_t}(\theta^{(t)}, \phi^{(t)}).$$

Each update uses only B observations to estimate the full-data gradient.

Latent Monte Carlo

For each observed x_i, draw independent latent samples:

$$z_i^{(r)} \sim q_\phi(\cdot \mid x_i), \qquad r = 1, \dots, R.$$

Estimate the local ELBO by

$$\boxed{\widehat{\mathcal{L}}_i(\theta, \phi) = \frac{1}{R} \sum_{r=1}^{R} \left[\log p_\theta(x_i, z_i^{(r)}) - \log q_\phi(z_i^{(r)} \mid x_i) \right].}$$

For fixed x_i, θ, ϕ, this estimate is unbiased:

$$\mathbb{E}_{\mathrm{MC}}[\widehat{\mathcal{L}}_i(\theta, \phi)] = \mathcal{L}_i(\theta, \phi),$$

where the expectation is over the latent draws.

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We can estimate the ELBO. Can we obtain its gradient by differentiating this sample average?

A differentiation issue

Consider a Gaussian encoder:

$$q_\phi(z \mid x_i) = \mathcal{N}(\mu_\phi(x_i), \Sigma_\phi(x_i)) \,.$$

The computation proceeds as

$$\phi \longrightarrow (\mu_\phi(x_i), \Sigma_\phi(x_i)) \xrightarrow{\text{sample}} z_i^{(r)}.$$

Where can differentiation break?
If the sampling operation returns $z_i^{(r)}$ as a **detached value**, automatic differentiation treats it as constant:

$$\nabla_\phi \log p_\theta(x_i, z_i^{(r)})\Big|_{z_i^{(r)} \text{ held fixed}} = 0.$$

Changing ϕ changes the distribution of the samples, but this effect is absent from the recorded gradient.

A differentiation issue

The encoder parameters ϕ affect both the integrand and the distribution used to sample Z:

$$\nabla_\phi \mathcal{L}_i(\theta, \phi) = \nabla_\phi \mathbb{E}_{q_\phi(z|x_i)} \left[\log p_\theta(x_i, Z) - \log q_\phi(Z \mid x_i)\right]$$

$$\neq \mathbb{E}_{q_\phi(z|x_i)} \left[\nabla_\phi \left\{\log p_\theta(x_i, Z) - \log q_\phi(Z \mid x_i)\right\}\right] \quad \text{in general.}$$

Why? Write $f_\phi(z) = \log p_\theta(x_i, z) - \log q_\phi(z \mid x_i)$. The product rule gives

$$\nabla_\phi \mathcal{L}_i = \mathbb{E}_{q_\phi}[\nabla_\phi f_\phi(Z)] + \underbrace{\int f_\phi(z)\, \nabla_\phi q_\phi(z \mid x_i)\, dz}_{\text{effect of changing the sampling distribution}} .$$

Treating the sampled $z_i^{(r)}$ as constants omits this additional term.

Reparameterization trick

For a Gaussian encoder, write $\Sigma_\phi(x) = L_\phi(x)L_\phi(x)^\top$ and sample

$$\boxed{Z = \mu_\phi(x) + L_\phi(x)\epsilon = g_\phi(\epsilon, x), \qquad \epsilon \sim \mathcal{N}(0, I_K).}$$

The distribution of ϵ is independent of ϕ.

Expressing the expectation using this fixed noise distribution,

$$\begin{aligned}\nabla_\phi \mathbb{E}_{q_\phi(z|x)}[f(Z)] &= \nabla_\phi \mathbb{E}_{p(\epsilon)}[f(g_\phi(\epsilon, x))] \\ &= \mathbb{E}_{p(\epsilon)}\left[\nabla_\phi f(g_\phi(\epsilon, x))\right].\end{aligned}$$

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Monte Carlo gradient estimator: draw $\epsilon^{(r)} \sim \mathcal{N}(0, I_K)$ independently and compute

$$\boxed{\nabla_\phi \mathcal{L} \approx \frac{1}{R}\sum_{r=1}^{R} \nabla_\phi f(g_\phi(\epsilon^{(r)}, x)).}$$

Blackbox variational inference

Fix x and θ. Write $q_\phi(z) = q_\phi(z \mid x)$ and define

$$f_\phi(z) = \log p_\theta(x, z) - \log q_\phi(z), \qquad \mathcal{L}(\theta, \phi) = \int q_\phi(z) f_\phi(z)\, dz.$$

Differentiate using the product rule:

$$\begin{aligned}
\nabla_\phi \mathcal{L} &= \int f_\phi(z) \nabla_\phi q_\phi(z)\, dz + \int q_\phi(z) \nabla_\phi f_\phi(z)\, dz \\
&= \mathbb{E}_{q_\phi}[\{f_\phi(Z) - 1\} \nabla_\phi \log q_\phi(Z)] \\
&= \boxed{\mathbb{E}_{q_\phi}[f_\phi(Z) \nabla_\phi \log q_\phi(Z)]}.
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\end{aligned}$$

Monte Carlo gradient estimator: draw $z^{(r)} \sim q_\phi$ independently and compute

$$\boxed{\nabla_\phi \mathcal{L} \approx \frac{1}{R} \sum_{r=1}^{R} f_\phi(z^{(r)})\, \nabla_\phi \log q_\phi(z^{(r)}).}$$

Comparing two solutions

1. Blackbox variational inference (BBVI): score-function gradients

- Applicable to both discrete and continuous latent variables
- Gradient estimates can have high variance.

2. Reparameterization: pathwise gradients

- Particularly convenient for Gaussian encoders, but might not applicable in general.
- Gradient estimates have relative smaller variance.

Variational Bayes

In Bayesian analysis, unknown model parameters θ also receive a prior:

$$p(x, z, \theta) = p(x \mid z, \theta)\, p(z)\, p(\theta).$$

We approximate the joint posterior of parameters and latent variables:

$$q(\theta, z) \approx p(\theta, z \mid x).$$

The same ELBO construction applies:

$$\boxed{\mathcal{L}(q) = \mathbb{E}_q[\log p(x, Z, \Theta)] - \mathbb{E}_q[\log q(\Theta, Z)].}$$

$$\log p(x) = \mathcal{L}(q) + \mathrm{KL}\{q(\theta, z) \| p(\theta, z \mid x)\}.$$

Example 3

A Bayesian Gaussian mixture:
Consider n real-valued observations and K components:

$$\mu_k \sim \mathcal{N}(0, \tau^2), \qquad k = 1, \ldots, K,$$

$$c_i \sim \text{Categorical}(1/K, \ldots, 1/K), \qquad i = 1, \ldots, n,$$

$$X_i \mid c_i = k, \mu \sim \mathcal{N}(\mu_k, 1).$$

- **Global unknowns:** component means $\mu = (\mu_1, \ldots, \mu_K)$.
- **Local unknowns:** assignments $c_i \in \{1, \ldots, K\}$.
- **Fixed quantities:** treat prior variance τ^2 as hyperparameter and is fixed in this example.

The mean-field family

Collect the unknown quantities into blocks $w = (w_1, \ldots, w_M)$.
The **mean-field variational family** assumes

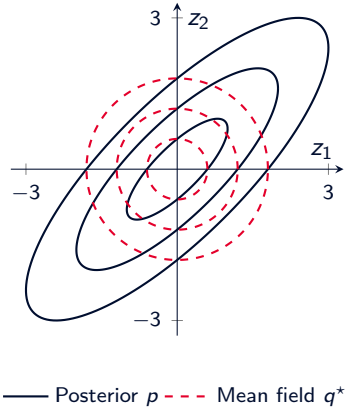
$$\boxed{q(w) = \prod_{j=1}^{M} q_j(w_j).}$$

- Independence is imposed on the approximation q.
- Each block may be a scalar or a vector.

Factorization simplifies optimization, but excludes posterior dependence between blocks.

What mean field loses

For two latent variables, mean field imposes $q(z_1, z_2) = q_1(z_1)q_2(z_2)$.



Correlated posterior

$$p(z \mid x) = \mathcal{N}\left(0, \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}\right).$$

Mean-field optimum

Minimizing $\mathrm{KL}(q\|p)$ gives

$$q^\star(z) = \mathcal{N}(0, 0.36 I_2).$$

Mean field removes dependence.

Mean field for the mixture

Use one factor for each mean and each assignment:

$$q(\mu, c) = \prod_{k=1}^{K} q_{\mu_k}(\mu_k) \prod_{i=1}^{n} q_{c_i}(c_i).$$

Choose Gaussian and categorical factors:

$$q_{\mu_k}(\mu_k) = \mathcal{N}(m_k, s_k^2), \qquad q_{c_i}(c_i = k) = r_{ik},$$

where

$$s_k^2 > 0, \qquad r_{ik} \geq 0, \qquad \sum_{k=1}^{K} r_{ik} = 1.$$

Optimize $(\theta, \phi_{[1:n]})$, where $\theta = (m_{[1:K]}, s^2_{[1:K]})$, and $\phi_i = r_{i[1:K]}$.

Coordinate ascent VI

Coordinate ascent variational inference (CAVI) updates one factor while holding the others fixed.

Write

$$q_{-j}(w_{-j}) = \prod_{\ell \neq j} q_\ell(w_\ell).$$

The coordinate-optimal update is

$$\boxed{q_j^{\text{new}}(w_j) \propto \exp\left\{\mathbb{E}_{q_{-j}}[\log p(x, w_j, W_{-j})]\right\}.}$$

1. Initialize the variational factors.
2. Cycle through $j = 1, \ldots, M$, updating and normalizing each factor using the current values of the others.
3. Repeat until the ELBO stabilizes.

Why the CAVI update works

Write ELBO:

$$\mathcal{L}(q) = \mathbb{E}_q \log p(x, W) - \mathbb{E}_q \log q(W)$$

Hold q_{-j} fixed and define

$$h_j(w_j) = \mathbb{E}_{q_{-j}}[\log p(x, w_j, W_{-j})].$$

Write the ELBO as a function of q_j:

$$\begin{aligned}\mathcal{L}(q_j) &= \mathbb{E}_{q_j}\mathbb{E}_{q_{-j}}[\log p(x, W_j, W_{-j})] - \mathbb{E}_{q_j} \log q_j(W_j) - \mathbb{E}_{q_{-j}} \log q_{-j}(W_{-j}) \\ &= \mathbb{E}_{q_j} h_j(W_j) - \mathbb{E}_{q_j} \log q_j(W_j) + \text{const.}\end{aligned}$$

Why the CAVI update works

Recall that:

$$\mathcal{L}(q_j) = \mathbb{E}_{q_j} h_j(W_j) - \mathbb{E}_{q_j} \log q_j(W_j) + \text{const.}$$

Define the normalized density

$$\widetilde{q}_j(w_j) = \frac{\exp\{h_j(w_j)\}}{\mathcal{Z}_j}, \qquad \mathcal{Z}_j = \int \exp\{h_j(u)\}\, du.$$

Substitution gives

$$\boxed{\mathcal{L}(q_j) = -\,\mathrm{KL}(q_j \| \widetilde{q}_j) + \log \mathcal{Z}_j + \text{const.}}$$

Thus the maximum occurs at $q_j = \widetilde{q}_j$. Therefore

$$q_j^{\text{new}}(w_j) \propto \exp\Big\{\mathbb{E}_{q_{-j}}[\log p(x, w_j, W_{-j})]\Big\}.$$

Example 3 continued

Define the assignment indicator

$$d_{ik} = \mathbf{1}\{c_i = k\}.$$

For the Bayesian Gaussian mixture, the **joint density** is

$$p(\mu, c, x) = \prod_{k=1}^{K} p(\mu_k) \prod_{i=1}^{n} p(c_i) \prod_{i=1}^{n} \prod_{k=1}^{K} p(x_i \mid c_i, \mu_k)^{d_{ik}}$$

Taking logarithms gives

$$\boxed{\log p(\mu, c, x) = -\frac{1}{2\tau^2} \sum_{k=1}^{K} \mu_k^2 - \frac{1}{2} \sum_{i=1}^{n} \sum_{k=1}^{K} d_{ik}(x_i - \mu_k)^2 + \text{const.}}$$

Updating the assignments

Hold the Gaussian factors fixed. For a candidate assignment $c_i = k$,

$$\log q_{c_i}^{\text{new}}(k) = -\frac{1}{2}\mathbb{E}_{q_{\mu_k}}[(x_i - \mu_k)^2] - \frac{1}{2}\sum_{j \neq i}\sum_{k=1}^{K}\mathbb{E}_{q_{c_j} q_{\mu_k}}[d_{jk}(x_j - \mu_k)^2]$$

$$- \frac{1}{2\tau^2}\sum_{k=1}^{K}\mathbb{E}_{q_{\mu_k}}[\mu_k^2] + \text{const.}$$

$$= -\frac{1}{2}\mathbb{E}_{q_{\mu_k}}[(x_i - \mu_k)^2] + +\text{const.}$$

Dropping the common term $-x_i^2/2$, define

$$a_{ik} = x_i m_k - \frac{1}{2}(m_k^2 + s_k^2).$$

Normalize across components:

$$\boxed{r_{ik}^{\text{new}} = \frac{\exp(a_{ik})}{\sum_{\ell=1}^{K}\exp(a_{i\ell})}.}$$

Updating component means

Hold the assignment probabilities fixed and define

$$N_k = \sum_{i=1}^{n} r_{ik}, \qquad T_k = \sum_{i=1}^{n} r_{ik} x_i.$$

The terms involving μ_k give

$$\log q_{\mu_k}^{\text{new}}(\mu_k) = -\frac{\mu_k^2}{2\tau^2} - \frac{1}{2}\sum_{i=1}^{n} r_{ik}(x_i - \mu_k)^2 + \text{const.}$$

$$= -\frac{1}{2}(\tau^{-2} + N_k)\mu_k^2 + T_k \mu_k + \text{const.}$$

This is a Gaussian log density. Completing the square yields

$$q_{\mu_k}^{\text{new}}(\mu_k) = \mathcal{N}(m_k^{\text{new}}, s_k^{2,\text{new}}),$$

with

$$\boxed{s_k^{2,\text{new}} = \frac{1}{\tau^{-2} + N_k}, \qquad m_k^{\text{new}} = \frac{T_k}{\tau^{-2} + N_k}.}$$

The mixture ELBO

All terms in $\mathcal{L}(q) = \mathbb{E}_q[\log p(\mu, c, x)] + \mathcal{H}(q)$ can be evaluated analytically:

$$\begin{aligned}\mathcal{L}(q) = & -\frac{n}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{n}\sum_{k=1}^{K} r_{ik}\left\{(x_i - m_k)^2 + s_k^2\right\} \\ & -\frac{1}{2}\sum_{k=1}^{K}\left\{\frac{m_k^2 + s_k^2}{\tau^2} - 1 + \log\frac{\tau^2}{s_k^2}\right\} \\ & -\sum_{i=1}^{n}\sum_{k=1}^{K} r_{ik}\log(r_{ik}K).\end{aligned}$$

- First line: expected joint log likelihood.
- Second line: negative KL from the Gaussian factors to their priors.
- Third line: negative KL from the assignment factors to the uniform prior.

CAVI algorithm

Input: observations $x_{1:n}$, component count K, and prior variance τ^2.
Initialize: distinct means m_k and positive variances s_k^2.

Repeat until the ELBO stabilizes:

1. **Update assignments**, for every i:

$$a_{ik} = x_i m_k - \frac{1}{2}(m_k^2 + s_k^2), \qquad r_{ik} = \frac{\exp(a_{ik})}{\sum_{\ell=1}^{K} \exp(a_{i\ell})}.$$

2. **Update mean distributions**, for every k:

$$N_k = \sum_i r_{ik}, \qquad T_k = \sum_i r_{ik} x_i,$$

$$s_k^2 = \frac{1}{\tau^{-2} + N_k}, \qquad m_k = \frac{T_k}{\tau^{-2} + N_k}.$$

3. Compute the ELBO.

Return: $q_{\mu_k} = \mathcal{N}(m_k, s_k^2)$ and $q_{c_i}(k) = r_{ik}$.

Summary

The central identity is

$$\boxed{\log p_\theta(x) = \mathcal{L}(q, \theta) + \mathrm{KL}\{q(z) \| p_\theta(z \mid x)\}.}$$

1. **VI turns posterior approximation into optimization.**
   For fixed θ, maximizing the ELBO minimizes posterior KL.

2. **Different tools address different computational costs.**
   AVI: learn a shared inference rule.
   SVI: estimate full-data gradients using minibatches.
   MC: approximate latent-variable expectations using samples.

3. **Variational Bayes approximates parameter uncertainty.**
   Mean field factorizes the approximation and **CAVI** gives the optimization path.

References

1. Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). *Variational Inference: A Review for Statisticians*. Journal of the American Statistical Association, 112(518), 859–877.

2. Murphy, K. P. (2023). *Probabilistic Machine Learning: Advanced Topics*. MIT Press.

3. Kingma, D. P., & Welling, M. (2014). *Auto-Encoding Variational Bayes*. International Conference on Learning Representations (ICLR).

4. Ranganath, R., Gerrish, S., & Blei, D. (2014). *Black Box Variational Inference*. AISTATS, Proceedings of Machine Learning Research, 33, 814–822.

5. Hoffman, M. D., Blei, D. M., Wang, C., & Paisley, J. (2013). *Stochastic Variational Inference*. Journal of Machine Learning Research, 14(40), 1303–1347.

6. Kingma, D. P., & Welling, M. (2019). *An Introduction to Variational Autoencoders*. Foundations and Trends in Machine Learning, 12(4), 307–392.